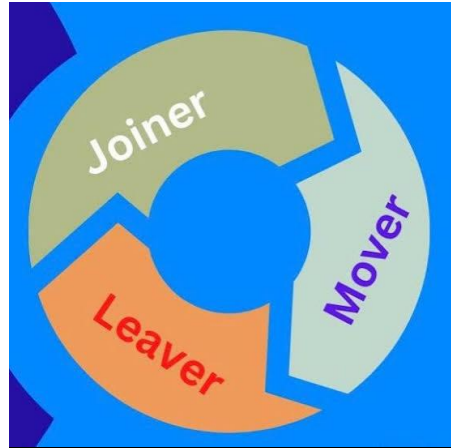




Joiner-Mover-Leaver (JML) Workflow

Identity Lifecycle Management using Microsoft Entra ID

1. Introduction

Identity Lifecycle Management is one of the most important components of Identity and Access Management (IAM). Organizations must ensure that every employee receives the correct level of access when joining the company, that permissions are updated when job responsibilities change, and that access is removed immediately when employment ends.

The Joiner-Mover-Leaver (JML) process provides a structured approach to managing digital identities throughout the employee lifecycle. In this project, Microsoft Entra ID is used to simulate real-world identity management practices by creating users, assigning groups and roles, implementing Role-Based Access Control (RBAC), and demonstrating user provisioning and de-provisioning.

2. What is Joiner-Mover-Leaver (JML)?

Joiner-Mover-Leaver (JML) is an Identity Governance process used to manage employee accounts throughout their lifecycle.

It consists of three stages:

- **Joiner:** Creating and provisioning accounts for new employees.
- **Mover:** Updating permissions when employees change roles or departments.
- **Leaver:** Removing access when employees leave the organization.

Proper implementation of JML reduces security risks, prevents unauthorized access, and supports Zero Trust security principles.

3. Objectives

The objectives of this JML workflow are to:

- Standardize identity lifecycle management.
- Ensure users receive only the minimum permissions required.
- Apply the Principle of Least Privilege.
- Reduce orphaned accounts.
- Improve compliance and security.
- Demonstrate practical Microsoft Entra ID administration.

4. Joiner Process (New Employee)

Scenario

A new employee named **Bilal** joins the Human Resources (HR) department.

The following actions are performed by the Identity Administrator.

Step 1 – Receive Employee Information

The HR department submits the following details:

- Employee Name
- Employee ID
- Department
- Job Title
- Manager
- Employment Type
- Office Location

Example:

Field	Value
Name	Bilal
Employee ID	EMP005

Field	Value
Department	HR
Job Title	HR Executive
Manager	Umer
Employment Type	Full-Time
Office	Islamabad

Step 2 – Create User Account

Create the employee account in Microsoft Entra ID.

Information configured:

- Display Name
- User Principal Name (UPN)
- Password
- Account Status (Enabled)

Example:

Display Name: Bilal

Account:Enabled

Step 3 – Assign Security Group

The employee is added to the appropriate department group.

Example:

HR Group

Benefits:

- Easier permission management
- Simplified administration
- Consistent access

Step 4 – Assign Default Role

Bilal receives only the **Default User** role.

No administrative privileges are assigned.

This follows the Principle of Least Privilege.

Step 5 – Assign Applications

The employee receives access only to required applications.

Examples:

- Microsoft Outlook
- Microsoft Teams
- SharePoint
- Internal HR Portal

Step 6 – Enable MFA

The employee registers:

- Microsoft Authenticator
- Phone Number
- Backup Authentication Method

(In this lab, MFA is documented but not implemented because Microsoft Entra ID Free does not support advanced MFA policies.)

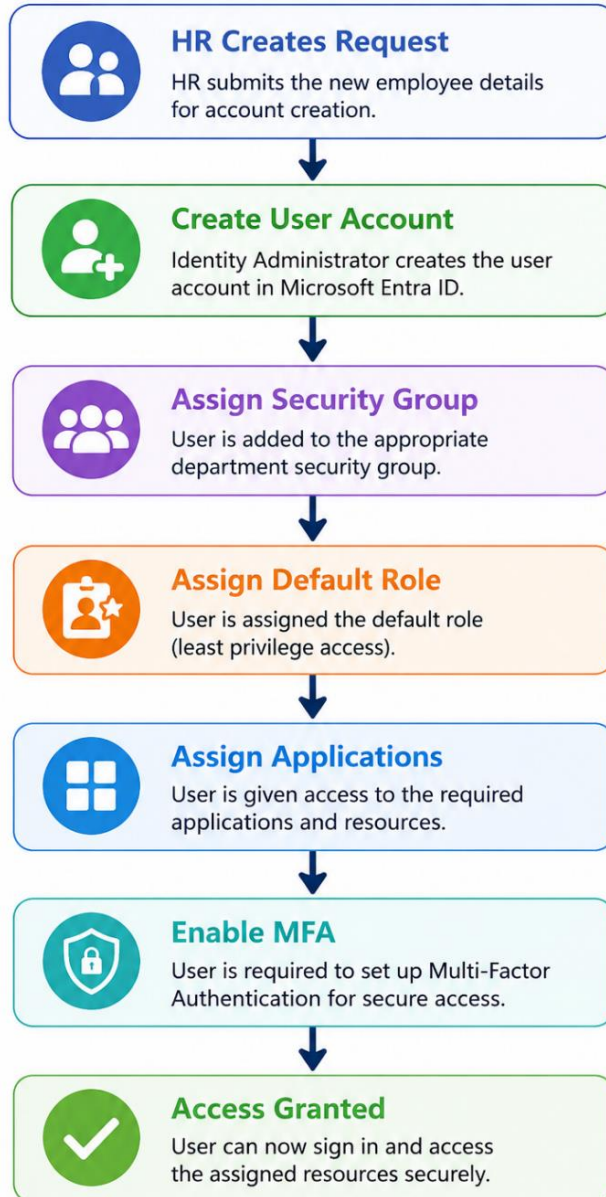
Step 7 – Verify Access

Confirm:

- User can sign in
- Group membership is correct
- Applications are accessible
- No unnecessary permissions exist

Joiner Workflow:

— Joiner Workflow —



5. Mover Process (Department Change)

Scenario

Ahmed transfers from the Contractors department to the IT department.

His permissions must be updated.

Step 1 – Receive Transfer Request

HR informs IT that Ahmed has changed departments.

Step 2 – Remove Previous Access

Remove Ahmed from:

Contractors Group

Remove permissions related to contractor resources.

Step 3 – Add New Group

Add Ahmed to:

IT Group

Step 4 – Update Role

If required:

Assign

Security Administrator

Otherwise keep him as a normal IT user.

Step 5 – Grant New Permissions

Ahmed receives access to:

- IT Resources
- Security Dashboards
- Required Applications

Step 6 – Review Existing Access

Check:

- Previous permissions removed
- New permissions applied
- Least Privilege maintained

Mover Process:

Mover Workflow



6. Leaver Process (Employee Leaves)

Scenario

Ali resigns from CyberSecurity Lab.

His account must be removed securely.

Step 1 – Receive Exit Notification

HR confirms employee resignation.

Step 2 – Disable Account

Immediately prevent user login.

This stops unauthorized access.

(In this lab, the user account was deleted because the free tenant does not include every lifecycle feature.)

Step 3 – Revoke Active Sessions

Terminate all active sign-in sessions.

Step 4 – Remove Group Membership

Remove the employee from:

- HR
- IT
- Contractors
- Managers

Step 5 – Remove Administrative Roles

Remove:

- Helpdesk Administrator
- Security Administrator

if assigned.

Step 6 – Remove Licenses

Remove Microsoft licenses.

(Not demonstrated because no Microsoft 365 licenses were assigned in this lab.)

Step 7 – Delete User

Delete the user account from Microsoft Entra ID.

Step 8 – Record Offboarding

Document:

- Date
- Reviewer
- Reason
- Completion Status

Leaver Workflow:

— Leaver Workflow —



7. Access Review

Access Reviews ensure users still require their assigned permissions.

During the review:

- Verify group memberships
- Review administrative roles
- Check inactive users
- Remove excessive permissions

Example Review Table

User	Group	Role	Decision
Umer	Managers	Helpdesk Administrator	Keep
Asad	IT	Security Administrator	Keep
Ahmed	Contractors	User	Keep
Bilal	HR	User	Keep
Ali	HR	User	Removed

8. Roles and Responsibilities

Role	Responsibility
HR Department	Submit join, transfer, and exit requests
Identity Administrator	Create, modify, and delete accounts
Security Administrator	Review permissions and security settings
Helpdesk Administrator	Reset passwords and unlock accounts
Manager	Approve department access
Global Administrator	Manage the Microsoft Entra ID tenant

9. Security Best Practices

- Apply the Principle of Least Privilege.
- Use Security Groups instead of assigning permissions directly to users.
- Review permissions regularly.
- Remove unnecessary administrator roles.
- Delete inactive accounts promptly.
- Use strong passwords.
- Enable MFA whenever licensing allows.
- Document all identity changes.

- Monitor sign-in and audit logs.

10. Benefits of JML

- Improves organizational security.
- Prevents unauthorized access.
- Reduces insider threats.
- Simplifies user management.
- Supports compliance requirements.
- Enhances identity governance.
- Ensures consistent access control.
- Supports Zero Trust architecture.

11. Conclusion

The Joiner-Mover-Leaver workflow implemented in this Microsoft Entra ID lab demonstrates how user identities can be securely managed throughout the employee lifecycle. The project covered user provisioning, group assignments, role-based access control, role changes, and user offboarding while following the Principle of Least Privilege. Although advanced features such as automated lifecycle workflows and Conditional Access require Microsoft Entra ID Premium licensing, the lab successfully demonstrates the core concepts and administrative processes used in enterprise identity and access management environments.